

ABDUL RAHIMAN MIHAD

AI Engineer | LLM Agents and Real-Time Voice AI | Full-Stack and Cloud | MLOps

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Indian national | UAE visit visa | Available immediately

PROFESSIONAL SUMMARY

AI Engineer building **Vendoor**, a B2B supplier discovery and tender platform for the UAE F&B market, owning the product end to end from four production LLM agents and WhatsApp and email automation to a FastAPI and PostgreSQL backend on AWS. Previously built **Alpha**, a full-duplex real-time AI voice interviewer at HireUp, lifting navigation accuracy from about 80% to 100%. Strong in Generative AI, agentic LLM systems, RAG, NLP, Computer Vision and MLOps. Microsoft Azure certified.

CORE COMPETENCIES

Artificial Intelligence (AI) · Machine Learning (ML) · Generative AI · Large Language Models (LLM) · LLM Agents and Tool Use · Natural Language Processing (NLP) · Computer Vision · Conversational AI · Voice AI · RAG · Vector Search · Prompt Engineering · LLM Fine-Tuning · Backend Architecture · Cloud Architecture (AWS) · MLOps

WORK EXPERIENCE

Full-Stack AI Engineer | Vendoor

Jul 2026 to Present

B2B supplier discovery and tender platform for the UAE F&B market · UAE · Early-stage startup, founder-led team

- **Built the platform end to end**, from the buyer-facing product and founder console to the AI pipeline, messaging and cloud infrastructure, taking it from MVP to production.
- **Developed four production LLM agents** with enforced structured output that turn buyer specs, spreadsheets and 5,000-SKU supplier catalogs into line items and parse free-text supplier quotes with per-line confidence.
- **Automated supplier outreach and quote collection** over the WhatsApp Cloud API and email, with HMAC-verified webhooks and atomic ingestion so retries never duplicate a quote and no message waits on a human.
- **Built the pricing normalisation engine and semantic catalog matching** with Voyage AI embeddings on pgvector, making every supplier price comparable at the buyer's quantity and matching aubergine to eggplant.
- **Migrated the backend** from Node.js and MySQL to Python 3.12, FastAPI and PostgreSQL on AWS (App Runner, ECS Fargate, RDS, S3, SQS), porting all 86 API routes under a CI-enforced contract with a zero-downtime cut-over.

AI/ML Engineer | HireUp

May 2026 to Jul 2026

Technology company headquartered in Japan (remote) · Built Alpha, a real-time AI voice interviewer · app.hireup.now

- **Designed the full-duplex real-time voice pipeline** with WebSocket streaming, concurrent asyncio loops, Deepgram Flux STT, WebRTC VAD and streaming TTS, **cutting startup latency by about 1.5 seconds**.
- **Raised interview navigation accuracy from about 80% to 100%** by replacing non-deterministic LLM routing with a deterministic resolver, verified across live interviews and automated tests.
- **Significantly cut LLM operating cost** by redesigning the conversation turn loop to remove redundant model calls from every turn.
- **Built an automated evaluation harness** that replays full interviews in seconds, with MLflow and LangSmith tracing to catch regressions across releases.

KEY PROJECTS

Self-Corrective RAG System · LangGraph, AWS Bedrock, FastAPI, MLflow, Docker, MCP · [Live Demo](#)

- **Reduced hallucinations by 40% and reached 92% accuracy** with a self-correcting multi-agent pipeline (document grading, hallucination checks, query rewriting) on Llama 3.1 70B via AWS Bedrock, with sub-1.5s responses.

End-to-End Litter Detection System · YOLOv8, PyTorch, OpenCV, CVAT, DVC, MLflow, Streamlit · [Live Demo](#)

- **Improved small-object detection accuracy (mAP) by 15%** on 1,500+ hand-annotated images across 50+ tracked experiments, deployed as a live Streamlit app on a reproducible DVC and MLflow pipeline.

Character-Level Language Model from Scratch · PyTorch, NumPy, custom autograd · [GitHub](#)

- Implemented a character-level language model and a **custom automatic differentiation engine** from first principles to understand how modern LLMs train and generate text.

EDUCATION

Bachelor of Computer Applications (BCA) | **Yenepoya University**, India Aug 2022 to Nov 2025
Specialised in Machine Learning, AI and Robotics · Industry-partnered programme with TCS and Microsoft

CERTIFICATIONS

Microsoft	Azure AI Fundamentals (AI-900) · Azure Fundamentals (AZ-900) · Data Fundamentals (DP-900)
IBM (Coursera)	Advanced LLM Fine-Tuning · Fine-Tuning Transformers · AI Agents with RAG and LangChain · LLM Architecture and Data Prep · Language Models with Transformers · Foundational Models for NLP · EDA for Machine Learning
Kaggle	Advanced SQL · Intro to SQL

TECHNICAL SKILLS

AI / LLM	Tool Use and Structured Output · RAG · Embeddings (Voyage AI, pgvector) · LangGraph · LangChain · Hugging Face · MCP · PEFT/LoRA · PyTorch · TensorFlow · NLP · Computer Vision
Real-Time AI	WebSockets · asyncio · Deepgram STT and TTS · WebRTC VAD · Web Audio API · Streaming Pipelines
Backend	Python 3.12 · FastAPI · SQLAlchemy · PostgreSQL · pgvector · Alembic · Node.js · Express · MySQL · Redis
Frontend	JavaScript · HTML · CSS · Tailwind · Streamlit
DevOps	Docker · GitHub Actions (OIDC) · CI/CD · pytest · ruff · MLflow · LangSmith · DVC · Model Monitoring
Cloud	AWS (App Runner, ECS Fargate, RDS, S3, SQS, Secrets Manager, Bedrock) · Azure · GCP · BigQuery
Integrations	Meta WhatsApp Cloud API and Flows · Resend · Supabase Auth · pypdf · openpyxl · fpdf2
Languages	English (Fluent) · Malayalam (Native) · Hindi (Elementary) · Arabic (Elementary)